

A Bird's Eye View of Android System Services

Android is gaining unprecedented momentum in the smartphone and tablet market. This huge success is usually attributed to factors like the open ecosystem, the choice of Java as the application programming language, the choice of the Linux kernel, etc. A lesser known fact is the existence of an application framework that has feature-rich APIs, which help in developing 'cool' applications in no time, besides being customisable to a great extent. At the heart of the Android framework are the 'system services' that manage almost everything in the Android platform. This article attempts to provide a bird's eye view of the 'system services' in Android.



In Android applications, 'services' are typically used to perform background operations that take a considerable amount of time. This ensures faster responsiveness to the 'main' thread (a.k.a. the UI thread) of an application, with which the user is directly interacting. The life cycle of the 'services' used in applications is managed by the Android Framework, i.e., these services have 'startService()', 'bindService()' and 'stopService()' calls that are called when an activity (or some other component) starts, binds or stops a service. The Android system might force-stop a service when available memory is low; <http://developer.android.com/guide/components/services.html> gives a detailed description of the Android 'service', which is an application component.

System services

The 'system' services play a key role in exposing the low-level functions of the hardware and the Linux kernel to the high-level applications. The 'system' services live from boot to reboot, i.e., the entire life of the system. There are about 70 system services in the Jelly Bean release of Android. Table 1 shows a list of some of the system services, the names of which are self-explanatory. It would easily take pages to

explain the functionality of each one of these, which is beyond the scope of this article.

ActivityManagerService (~14500 LOC)	LightsService
AlarmManagerService	NetworkManagementService
BackupManagerService	NotificationManagerService
BatteryService	StatusBarManagerService
ConnectivityService	VibratorService
CountryDetectorService	PackageManagerService (~10500 LOC)
DevicePolicyManagerService	PowerManagerService (~2500 LOC)
DeviceStorageMonitorService	WifiService
LocationManagerService	WindowManagerService (~11000 LOC)

Table 1: A list of some Android system services
LOC stands for Lines of code